

Conventional panel mounting type

KS53 Solid State Relay

Mini size and Single channel control type



Max AC load **25A**



Round screw terminal and fasten terminal is optional



Transparent finger-proof cover is available, compliant with **IP20** standard



Application Scenarios:

Mainly used in equipment such as rubber and plastics, commercial kitchen appliances, instant hot water dispensers, and coffee machines.



DESCRIPTION

KS53 is an AC output panel mount type SSR. It offers 4~32VDC input voltage option and two AC output voltage options 240VAC and 380VAC as well as two output current ratings 10A, and 25A for selection. The SSR provides photoelectric isolation between input and output with dielectric strength 4000V. KS53 has the built-in RC snubber circuit and is epoxy resin encapsulated.

- Built-in RC snubber circuit
- TRIAC output
- Removable protective cover
- Photoelectric isolation
- Dielectric strength 4000V

SELECTION GUIDE

KS53/	D-	24	Z	10	-L	Q	(XXX)
Type	Control voltage	Load voltage	Switching mode	Load current	LED indicator	Termination	Customer special code
	D:4~32 VDC	24:240V 38:380V	Z:Zero-cross	10:10A 25:25A	L:Included	Q:Quick connection Nil:Screw	

■ SELECTION GUIDE (Ta=25°C)

Control voltage range	(4 ~ 32)VDC
Must turn-on voltage	4VDC
Must turn-off voltage	1VDC
Max. input current	25mA
Max. reverse protection voltage	-32VDC

■ OUTPUT SPECIFICATIONS (Ta=25°C)

	D-□□Z10-L	D-□□Z25-L
Load voltage range	D-24□□	(48~280)VAC
	D-38□□	(48~440)VAC
Max. transient voltage	D-24□□	600Vpk
	D-38□□	800Vpk
Load current range (A)	0.1~10	0.1~25
Max. I ² t (10ms, A ² s)	72	312
Max. surge current (10ms)	120A _{pk}	250A _{pk}
Max. off-state leakage current	5mA	
Max. on-state voltage drop	1.5V _{r.m.s.}	
Max. turn-on time(Zero-cross type)	1/2 cycle + 1ms	
Max. turn-off time	1/2 cycle + 1ms	
Min. off-state dv/dt	200V/μS	

■ GENERAL SPECIFICATIONS (Ta=25°C)

Dielectric strength	2500VAC, 50~60Hz, 1min, input/output/base
	4000VAC, 50~60Hz, 1min, input/output
Insulation resistance	1000MΩ (500VDC)
Operating temperature	-30 ~ 80°C
Storage temperature	-30 ~ 100°C
Unit weight	Approx. 35g

■ PRECAUTIONS

1. Please pay special attention to the actual load current and the ambient temperature when doing the type selection. And the SSR requires proper heat sinking for heat dissipation in full load. For ambient temperature above 40°C, the load current must be derated. Please refer to the curve of Max. Load Current vs. Ambient Temperature for derating.
2. The heat produced by the SSR during the working process must be dissipated via the metal base of the SSR. Please coat the SSR metal base with some thermal grease or a thermal pad, and then firmly press the SSR against the heatsink to ensure the full adherence.

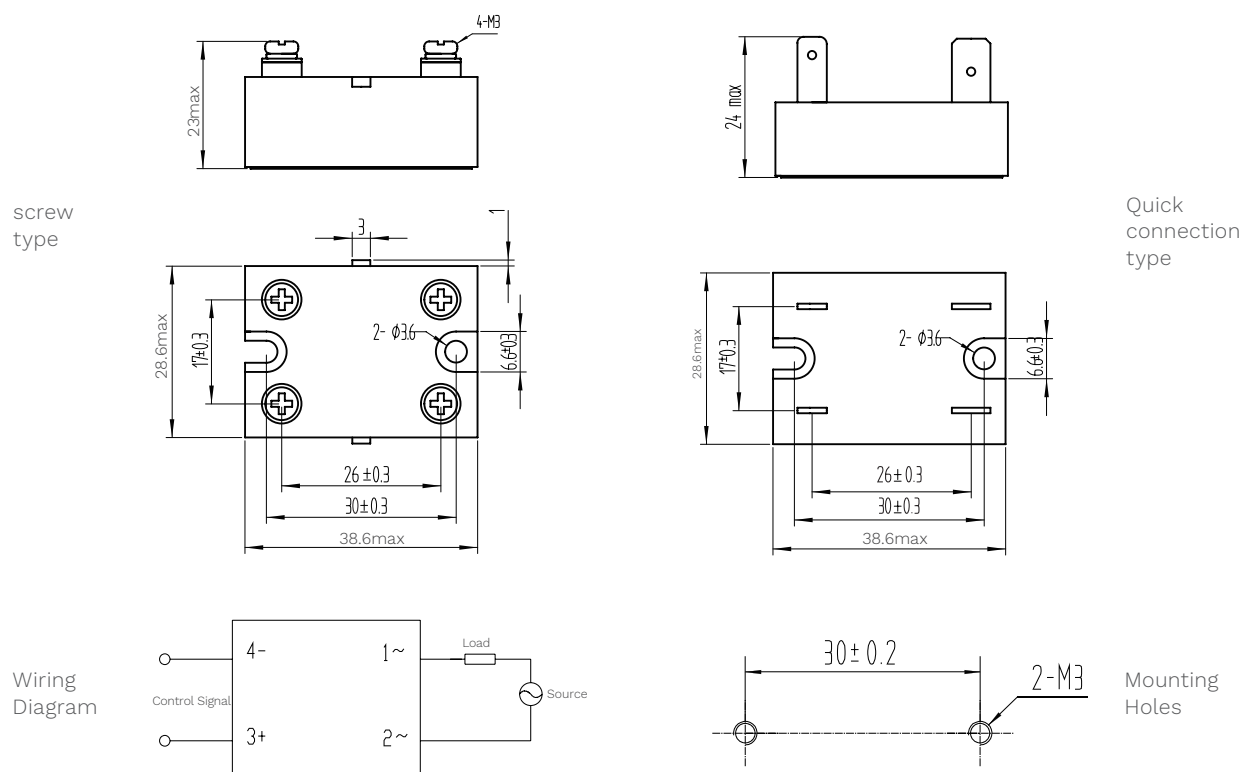
PRECAUTIONS

3. Tighten the SSR screw terminals properly. If the screws are loose, the SSR would be damaged by heat generated from connection. Also excessive screw mounting torque may damage the SSR's internal components. The recommended mounting torque range for M3 screw is 0.58~0.98N·m.
4. Please do not use the SSR exceeding the limitation which is specified on this datasheet.

OUTLINE DIMENSIONS, WIRING DIAGRAM

Unit: mm

Outline Dimensions



CHARACTERISTIC CURVES

